

Linear Regression, Revisited

STAT 200 – Introductory Statistics

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March 11, 2025



Today's Objective

After today's lecture, you should:

- Understand the theory behind testing the relationship between two numeric variables.
- Estimate a value of the dependent variable (and provide a confidence interval).
- Predict a value of the dependent variable (and provide a prediction interval).

Example #1: Violent Crime Rate in 1990 vs. 2000

Example

Understanding the persistence of violent crime in cities can help determine which cities are being successful in reducing it and which cities are not. What is the relationship between the violent crime rate in 1990 and in 2000?

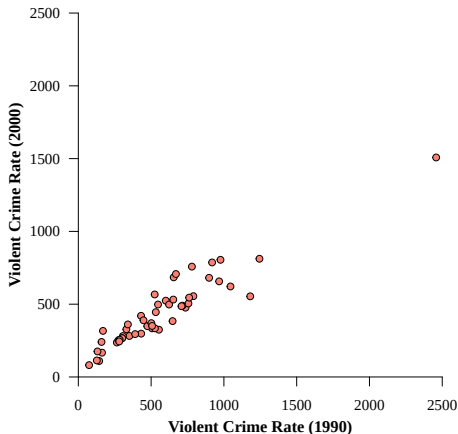
Note that we want to know *more* than just if there is a significant relationship.

We want to:

- specify how much one influences the other.
- estimate violent crime rates in 2000, given the value in 1990.
- Predict violent crime rates in 2000, given the value in 1990.

Example #1: Violent Crime Rate in 1990 vs. 2000

What is the relationship between these two variables?



To summarize the relationship, we use a “line of best fit”:

$$\hat{y} = \beta_0 + \beta_1 x$$

However, this raises the following question: What do we mean by the “best” fit?

The Theory

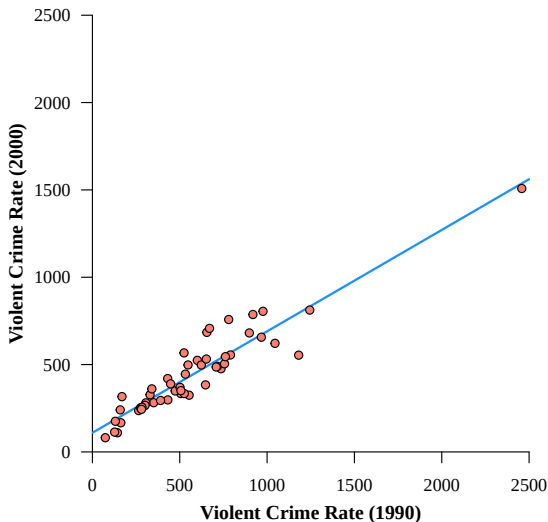
This is the most important question we can answer. Different answers lead to different fitting methods.

MATH/STAT 225 covers a few of the different methods. (There's still slots open in the winter term! Dr. Forsberg would love to have you!) It also covers the easiest method in great detail.

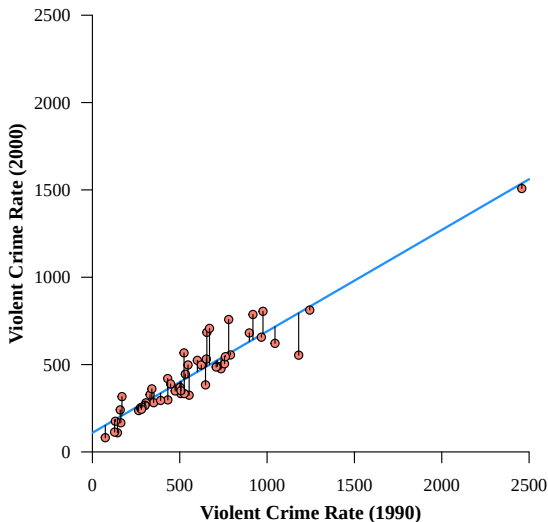
In this course, we will only look at the easiest method: ordinary least squares (OLS).

It is based on **minimizing the sum of the squared residual values**.

The Theory: The Line



The Theory: The Residuals



The Theory

Thus, our estimators of the y -intercept and slope (effect) are:

$$b_0 = \bar{y} - b_1\bar{x}$$
$$b_1 = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2}$$

By the way, the second formula can also be written as:

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$
$$= r \left(\frac{s_y}{s_x} \right)$$

This gives some insight into the slope, the correlation, and the relationship between them.

The Theory: Slope

It can be shown that the slope estimate has the following distribution:

$$b_1 \sim \text{NORMAL} \left(\beta_1, \frac{\sigma_e^2}{\sum (x_i - \bar{x})^2} \right)$$

With this distribution, we are able to formulate confidence intervals and p -values for hypotheses about the slope (the effect of x on y).

The proof of this is rather elementary and follows from the assumptions we make on the residuals. The three-line proof is provided in STAT 225. Also in that course is the proof of the confidence intervals and the p -values based on this distribution. (These are also three-line proofs.)

Note: Here, σ_e^2 is the population variance of the residuals.

The Theory: ANOVA

Sometimes, we want to draw conclusions about *the model, as a whole*.

To do this, we rely on ANOVA (does the model give information about the dependent variable?). As such, we need to calculate the following variations:

$$SSM = \sum_i (\hat{y}_i - \bar{y})^2 \quad \text{Explained by Model}$$

$$SSE = \sum_i (y_i - \hat{y}_i)^2 \quad \text{Remaining Error}$$

$$SST = \sum_i (y_i - \bar{y})^2 \quad \text{Total}$$

The Theory: ANOVA

The ANOVA table for the linear regression model with 1 independent variable:

Source	SS	df	MS	F	p
Model	$\sum_i (\hat{y}_i - \bar{y})^2$	1	$\frac{SSM}{1}$	$\frac{MSM}{MSE}$	$\mathbb{P}[F \geq f]$
Error	$\sum_i (y_i - \hat{y}_i)^2$	$n - 2$	$\frac{SSE}{n - 2}$		
Total	$\sum_i (y_i - \bar{y})^2$	$n - 1$			

The ANOVA table is useful for drawing conclusions about the model *as a whole*.

The Theory: ANOVA

Finally, we have a measure of how well the model fits the data. This is the so-called “ R -squared” value. There are two ways of calculating it:

$$R^2 = \frac{\text{Variation Explained}}{\text{Variation Remaining}} = \frac{SSR}{SST} = \frac{SST - SSE}{SST} = 1 - \frac{SSE}{SST},$$

and:

$$R^2 = r^2$$

Here, r is the correlation we learned about in the past.

Of course, for any real data set, it is unreasonable to do calculations by hand.

The calculations can be done in [R](#), of course:

- The first step is to fit the model.
- The second step is to summarize the results.

Returning to Example #1

This is the code to start to answer the questions raised at the start of this section:

```
### Preamble
source("http://rfs.kvasaheim.com/stat200.R")

# Load and attach the data
dt = read.csv("http://rfs.kvasaheim.com/data/crime.csv")
attach(dt)

### Create the model
modOLS = lm(vcrime00 ~ vcrime90)
```

Example #1: Violent Crime Rate in 1990 vs. 2000

To perform the model analysis, you need to run this line:

```
summary.aov(mod0LS)
```

Running that line produces the following output:

```
          Df Sum Sq Mean Sq F value Pr(>F)
vcrime90    1 2556238 2556238   349.3 <2e-16 ***
Residuals  49  358600    7318
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The Interpretation:

Because the p -value is less than our usual $\alpha = 0.05$, we reject the null hypothesis that the model does not offer significant understanding of the relationship between the violent crime rates in 1990 and 2000.

Example #1: Violent Crime Rate in 1990 vs. 2000

The ANOVA output tells us if the individual independent variables are statistically significant in describing the dependent variable. That is all the ANOVA output tells us.

If we want to determine things like the effect estimates, we use the “linear model” summary:

```
summary.lm(modOLS)
```

Side Note: If you fit using the `lm` function, the `summary` function is equivalent to the `summary.lm` function. That is, `summary(modOLS)` gives the same results.

Example #1: Violent Crime Rate in 1990 vs. 2000

Here is the resulting output. Make sure you can interpret the important parts.

Call:

```
lm(formula = vcrime00 ~ vcrime90)
```

Residuals:

Min	1Q	Median	3Q	Max
-241.32	-42.84	-18.04	40.97	208.41

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	109.52716	21.42679	5.112	5.27e-06 ***
vcrime90	0.58065	0.03107	18.689	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 85.55 on 49 degrees of freedom

Multiple R-squared: 0.877, Adjusted R-squared: 0.8745

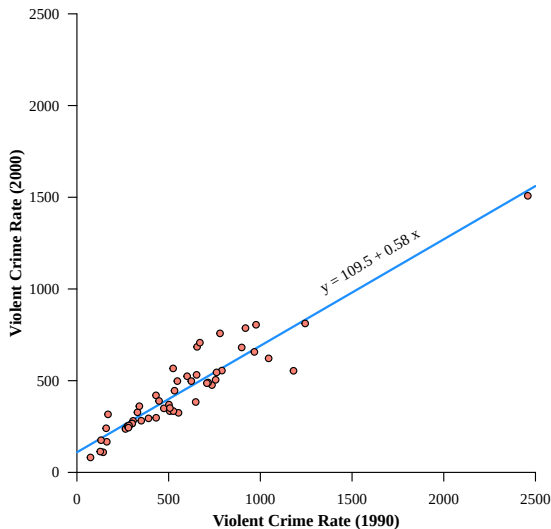
F-statistic: 349.3 on 1 and 49 DF, p-value: < 2.2e-16

Example #1: Violent Crime Rate in 1990 vs. 2000

From this output, we at least know the following:

- The effect of the 1990 violent crime rate on that of 2000 is 0.58065.
- For every increase in the violent crime rate in 1990 of 1, the estimated violent crime rate in 2000 increases by 0.58065.
- Those states with no violent crime in 1990 have an expected violent crime rate of 109.52716 in 2000.
- The effect of the 1990 violent crime rate on the 2000 violent crime rate is *not* 0 (p -value $\ll 0.0001$).

Example #1: Violent Crime Rate in 1990 vs. 2000



Example #1: Violent Crime Rate in 1990 vs. 2000

The previous code provides the point estimates for the intercept and slope. However, as we already know, we also need a confidence interval to communicate the precision of our estimates.

```
confint(modOLS)
```

	2.5 %	97.5 %
(Intercept)	66.4684173	152.5859064
vcrime90	0.5182157	0.6430849

Partial Conclusion:

We are 95% confident that the true effect of the 1990 violent crime rate on the 200 violent crime rate is between 0.518 and 0.643.

Example #1: Violent Crime Rate in 1990 vs. 2000

Since we can estimate a value of the dependent variable, we also need to calculate a confidence interval to indicate the precision of our estimate. Recall that the original problem had us estimate the 2000 violent crime rate for a state with a 1990 violent crime rate of 1500.

```
predict(modOLS, newdata=data.frame(vcrime=1500), interval="confidence")
```

	fit	lwr	upr
1	980.5026	917.7507	1043.255

Partial Conclusion:

We are 95% confident that the *expected* violent crime rate in 2000 for a state with a violent crime rate of 1500 in 1990 is between 918 and 1043, with a point estimate of 981.

Example #1: Violent Crime Rate in 1990 vs. 2000

Finally, we can predict a value of the dependent variable for the next state with that value of the independent variable.

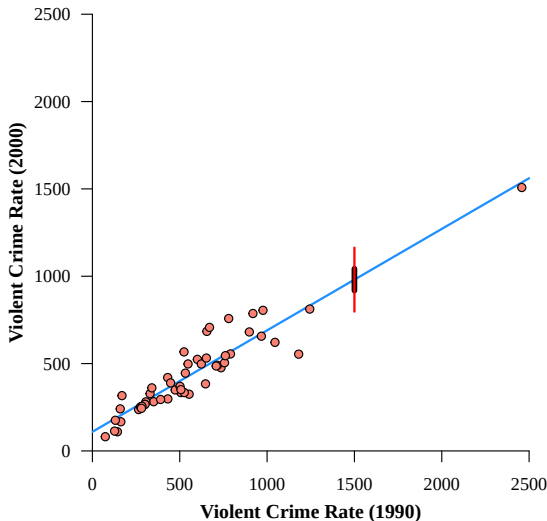
```
predict(modOLS, newdata=data.frame(vcrime=1500), interval="prediction")
```

	fit	lwr	upr
1	980.5026	797.4939	1163.511

Partial Conclusion:

We are 95% confident that the *actual* violent crime rate in 2000 for a state with a violent crime rate of 1500 in 1990 is between 797 and 1164, with a point estimate of 981.

Example #1: Violent Crime Rate in 1990 vs. 2000



Example #2: Violent Crime against Education

Example

What is the relationship between the violent crime rate in 2000 and the school enrollment in 1990?

The code for this analysis:

```
mod1 = lm(vcrime00 ~ enroll90)
```

Example #2: Property Crime Rate in 1990 vs. 2000

The resulting output is:

Call:

```
lm(formula = vcrime00 ~ enroll90)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-406.58	-160.05	-55.30	98.19	1005.73

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-211.613	804.596	-0.263	0.794
enroll90	7.095	8.732	0.813	0.420

Residual standard error: 242.3 on 49 degrees of freedom

Multiple R-squared: 0.01329, Adjusted R-squared: -0.006843

F-statistic: 0.6602 on 1 and 49 DF, p-value: 0.4204

Example #2: Property Crime Rate in 1990 vs. 2000

The *important* output is:

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-211.613	804.596	-0.263	0.794
enroll90	7.095	8.732	0.813	0.420

Example #2: Property Crime Rate in 1990 vs. 2000

Brief Conclusion: Because the p -value of 0.4204 is greater than our usual $\alpha = 0.05$, we cannot reject the null hypothesis. We did not detect a statistically significant relationship between the school enrollment in 1990 with the violent crime rate in 2000.

Example #3: Violent Crime against Wealth

Example

What is the relationship between the violent crime rate in 2000 and average wealth of the state in 1990? Also, what is the predicted 2000 violent crime rate for a state with average wealth \$50,000?

The code for this analysis:

```
mod2 = lm(vcrime00 ~ gspcap90)
summary(mod2)
```

```
confint(mod2)
```

```
predict(mod2, newdata=data.frame(gspcap90=50000), interval="prediction")
```

Example #3: Violent Crime against Wealth

The regression output is:

Call:

```
lm(formula = vcrime00 ~ gspcap90)
```

Residuals:

Min	1Q	Median	3Q	Max
-413.6	-123.9	-44.5	126.1	427.8

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	74.32013	88.97655	0.835	0.408
gspcap90	0.01598	0.00366	4.365	6.55e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 207 on 49 degrees of freedom

Multiple R-squared: 0.28, Adjusted R-squared: 0.2653

F-statistic: 19.06 on 1 and 49 DF, p-value: 6.545e-05

Example #3: Violent Crime against Wealth

Brief Conclusion: Because the p -value is much less than our usual $\alpha = 0.05$, we can reject the null hypothesis. We did detect a statistically significant relationship between the state's average wealth in 1990 with the violent crime rate in 2000.

	2.5 %	97.5 %
(Intercept)	-1.044849e+02	253.12520261
gspcap90	8.622233e-03	0.02333281

For every \$1,000 increase in average wealth in the state, the 2000 violent crime rate increase by an average of between 8.6 and 23.3.

	fit	lwr	upr
1	873.1962	408.6089	1337.784

We predict that a state with a GSP per capita of \$50,000 in 1990 will have a violent crime rate in 2000 between 409 and 1338, with a prediction of 873.

Example #4: Violent Crime against the Unemployment Rate

Example

What is the relationship between the violent crime rate in 2000 and the unemployment rate in 1990? If there is a relationship, estimate the 2000 violent crime rate given the 1990 unemployment rate is 10%.

The code for this analysis:

```
mod3 = lm(vcrime00 ~ unemp1990)
summary(mod3)
```


Example #3: Violent Crime against Wealth

The *important* output is:

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	293.51	168.04	1.747	0.087
unemp1990	27.06	30.09	0.899	0.373

Interpretation:

Because the p -value of 0.373 is greater than our usual $\alpha = 0.05$, we cannot reject the null hypothesis. We did not detect a statistically significant relationship between the unemployment rate in 1990 with the violent crime rate in 2000.

Beyond Simple Linear Regression

The previous example had one numeric dependent variable and *one* numeric independent variable, hence **Simple Linear Regression**. However, research usually posits that multiple variables affect the variable of interest (response variable). For example:

- state wealth depends on political, economic, and geographical factors.
- propensity for terrorist activity depends on minority concentration, economic strengths, and past activity.
- football power depends on the quality of the quarterback, the quality of the league, and the quality of the coach.
- success probability in STAT 200 depends on professor, preparation, study habits, and events happening in your life.

Nothing we have discussed so far can handle this many independent variables.

Violent Crime vs. Urban Percent *and* Region

Example

As a part of her research, a former Knox student hypothesized that there was a relationship between the violent crime rate and the percent of the population that is urban. She also hypothesized that this relationship differed across the four census regions.

Her actual research focused on policing policies across the United States and their effects on crime rate. This statistical work was just a small part of her effort. In fact, it was a very small part once she had the data.

From her literature review, she further hypothesized that there was a positive relationship between the violent crime rate and the urban percent, and that the relationship would be strongest in the Southern census region of the United States.

Violent Crime vs. Urban Percent *and* Region

The code for the *first* part of this analysis is:

```
mod1 = lm(vcrime00 ~ urbanp1990 * census4)
summary.aov(mod1)
```

Why `summary.aov`? There is a categorical independent variable and we want to determine if the *variable* and/or its *interaction* is important.

Violent Crime vs. Urban Percent *and* Region

The output from this part of the analysis is:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
urbanp1990	1	178031	178031	5.402	0.0249	*
census4	3	950724	316908	9.615	5.61e-05	***
urbanp1990:census4	3	368879	122960	3.731	0.0181	*
Residuals	43	1417204	32958			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note that the interaction between the two variables is significant ($p = 0.0181 < 0.05 = \alpha$). As such, both variables are important in the understanding of the violent crime rate in 2000 (in this model).

Furthermore, this result supports her second hypothesis, that the effect of urbanness on the violent crime rate varies from region to region.

Violent Crime vs. Urban Percent *and* Region

The important parts of the regression table are:

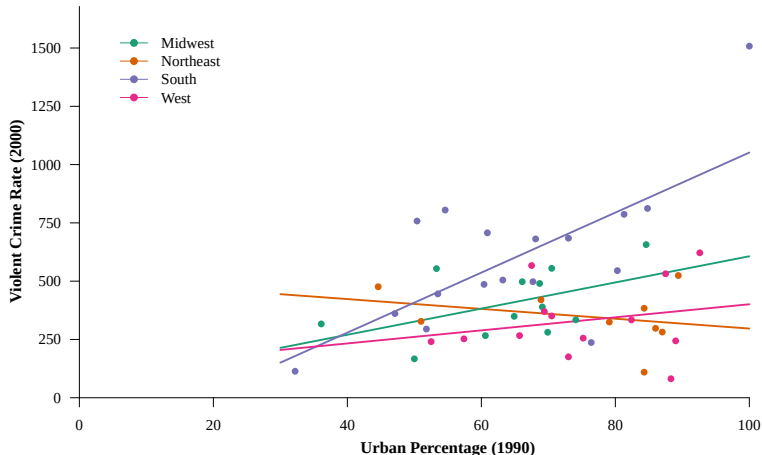
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	45.660	282.048	0.162	0.872
urbanp1990	5.611	4.331	1.296	0.202
census4Northeast	461.768	409.146	1.129	0.265
census4South	-281.398	336.494	-0.836	0.408
census4West	75.278	422.510	0.178	0.859
urbanp1990:census4Northeast	-7.718	5.809	-1.329	0.191
urbanp1990:census4South	7.263	5.124	1.417	0.164
urbanp1990:census4West	-2.811	6.003	-0.468	0.642

Note: Only three of the four regions are listed here. This is because all of the effects (estimates) are measured with respect to this “base” category (Midwest). Thus, the region effect of the South is 281.398 less than for the Midwest. Also, the interaction effect for the South is 7.263 greater than for the Midwest.

Violent Crime vs. Urban Percent *and* Region

Because the table on the previous slide is difficult to understand (for everyone), here is a graphic of the results.



Violent Crime vs. Urban Percent *and* Region

Because it is important to illustrate your results with specific examples, let us *estimate* the 2000 violent crime rate for a specific southern state with an urban population of 90%:

```
predict(mod1,  
        data.frame(urbanp1990=90, census4="South"),  
        interval="confidence")
```

fit	lwr	upr
922.8865	758.8959	1086.877

From these results, a 95% confidence interval for the 2000 violent crime rate for a southern state with a 90% urban population is from 759 to 1087, with a point estimate of 923 violent crimes per 100,000 people.

Violent Crime vs. Urban Percent *and* Region

Again, because it is important to illustrate your results with specific examples, let us predict the 2000 violent crime rate for a southern state with an urban population of 90%:

```
predict(mod1,  
        data.frame(urbanp1990=90, census4="South"),  
        interval="predict")
```

fit	lwr	upr
922.8865	521.7188	1324.054

From these results, a 95% prediction interval for the 2000 violent crime rate for a southern state with a 90% urban population is from 522 to 1324, with a point estimate of 923 violent crimes per 100,000 people.

Violent Crime vs. Property Crime and Region

Example

What is the relationship between the 2000 violent crime rate and the 1990 property crime rate, when controlling for the four census regions?

The code for the *first* part of this analysis is:

```
mod2 = lm(vcrime00 ~ pcrime90 * census4)
summary.aov(mod2)
```

Violent Crime vs. Property Crime and Region

The ANOVA output for this model is:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
pcrime90	1	1322873	1322873	52.352	5.88e-09	***
census4	3	423136	141045	5.582	0.00252	**
pcrime90:census4	3	82263	27421	1.085	0.36556	
Residuals	43	1086565	25269			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note that the interaction between the two variables is not statistically significant ($p = 0.36556 > 0.05 = \alpha$). As such, we need to modify our model by removing this interaction.

This decision arises from **Occam's Razor**: All things being equal, the simpler model is the more useful. This translates as one should *generally* remove non-significant variables.

Violent Crime vs. Property Crime and Region

This leads to the following model:

```
mod2a = lm(vcrime00 ~ pcrime90 + census4)
summary.aov(mod2a)
```

The ANOVA output for this *additive* model is:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
pcrime90	1	1322873	1322873	52.063	4.33e-09 ***
census4	3	423136	141045	5.551	0.00245 **
Residuals	46	1168828	25409		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Both variables have a detectable effect. As such, this is our model.

Violent Crime vs. Property Crime and Region

The important parts of the regression table are:

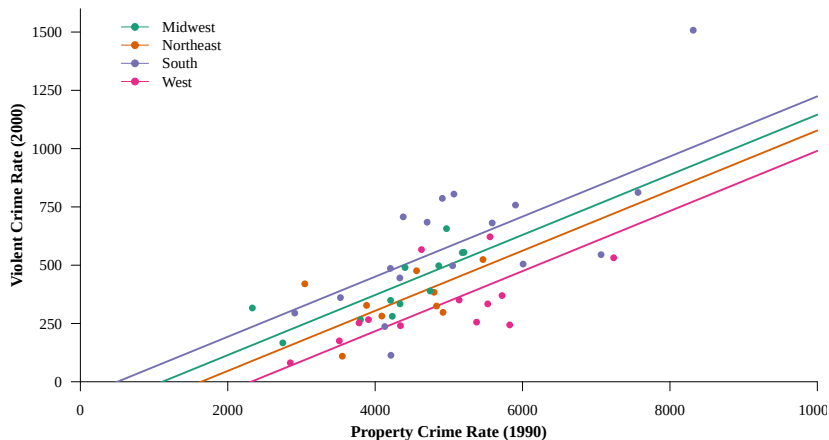
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-143.37481	96.65541	-1.483	0.1448	
pcrime90	0.12889	0.01999	6.448	6.15e-08	***
census4Northeast	-67.79721	70.31720	-0.964	0.3400	
census4South	78.62681	62.83648	1.251	0.2172	
census4West	-155.24230	65.02444	-2.387	0.0211	*

Again: Note that only three of the four regions are listed here. This is because all of the effects (estimates) are measured with respect to this “base” category (Midwest). Thus, for instance, the region effect of the South is 78.63 greater than for the Midwest.

Violent Crime vs. Property Crime and Region

The following is a graphic of the relationships between the property and the violent crime rate across the four census regions.



Violent Crime vs. Property Crime and Education Level

Example

What is the relationship between the 2000 violent crime rate and the 1990 property crime rate, when controlling for the level of education level of the state?

Note that this is a very different analysis than the previous ones. The two independent variables are *both* numeric. The analysis will be very similar, but the graphic will be quite different.

The code for starting this analysis is: The code for the *first* part of this analysis is:

```
mod3 = lm(vcrime00 ~ pcrime90 * waea90)
summary.aov(mod3)
```

Violent Crime vs. Property Crime and Education Level

The ANOVA output for this model is:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
pcrime90	1	1322873	1322873	44.273	2.74e-08	***
waea90	1	181846	181846	6.086	0.0173	*
pcrime90:waea90	1	5777	5777	0.193	0.6622	
Residuals	47	1404342	29880			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note that the interaction between the two variables is not statistically significant ($p = 0.6622 > 0.05 = \alpha$). As such, we need to modify our model by removing this interaction.

Violent Crime vs. Property Crime and Education Level

This leads to the following model:

```
mod3a = lm(vcrime00 ~ pcrime90 + waea90)
summary.aov(mod3a)
```

The ANOVA output for this *additive* model is:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
pcrime90	1	1322873	1322873	45.03	2.04e-08 ***
waea90	1	181846	181846	6.19	0.0164 *
Residuals	48	1410119	29377		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Both variables have a detectable effect. As such, this is our model.

Violent Crime vs. Property Crime and Education Level

The important parts of the regression table are:

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	555.94745	321.06193	1.732	0.0898	.
pcrime90	0.14060	0.02039	6.897	1.06e-08	***
waea90	-12.54722	5.04317	-2.488	0.0164	*

Interpretation: For every increase in the property crime rate of 1 in 100,000, the violent crime rate tends to increase by 0.14 points, all things being equal. Similarly, for every increase in the weighted average educational achievement of one point, the violent crime rate drops by 12.55 points.

Violent Crime vs. Property Crime and Education Level

So, how do we illustrate the relationship between/among the two independent *numeric* variables and the dependent variable? It is quite difficult.

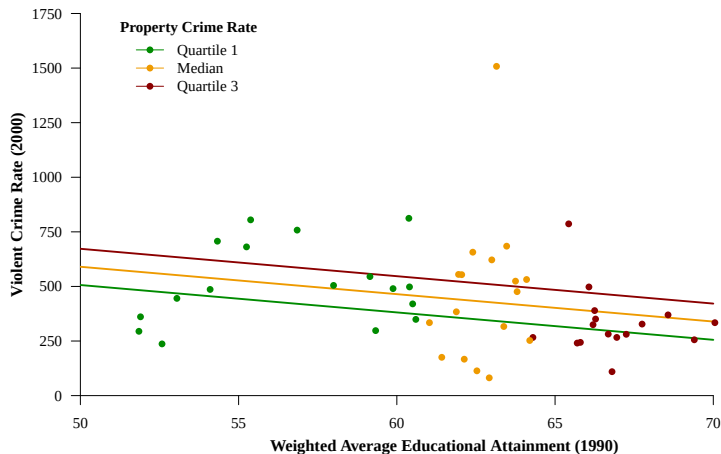
The key is to determine which variable *you want to emphasize* in your graphic. We shall designate this variable as the “*x*-variable”. Your *x*-variable will be assigned a range of values. All other variables need to be set to a specific value (or values).

Depending on how complicated your graphic, those specific values may be means, medians, or quartiles. This is where a conversation with others (and experience) is very useful. Remember that education is also a *community activity*. The best work is done with others.

Also, this is where acknowledging “good” is good enough is helpful. No graphic is perfect. They are (usually) good enough. They always have weaknesses, especially because different people look for different features in a graphic.

Violent Crime vs. Property Crime and Education Level

The following is a graphic of the relationships between the weighted average educational achievement and the violent crime rate for three exemplar property crime rates.



At this point in the course, we are *technically* able to model a Normally-distributed response variable...and that's all.

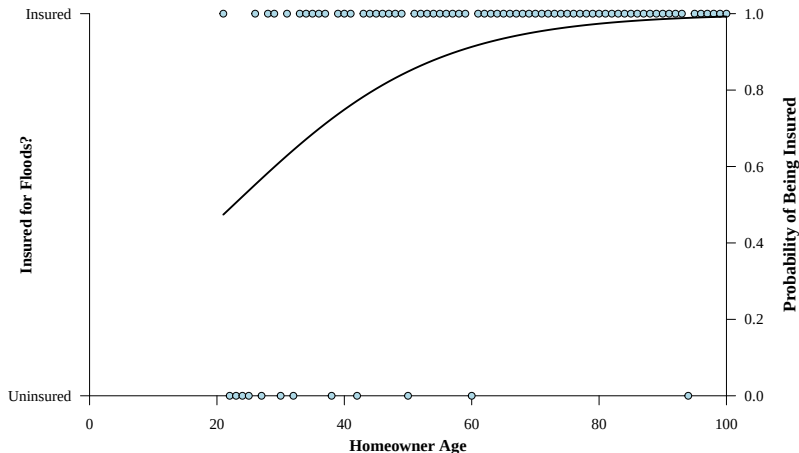
In the future, you may wish to model dependent variables that are not NORMAL, that are:

- just successes or failures (BERNOULLI).
- counts of successes over a specified number of trials (BINOMIAL).
- counts of successes over a specified time or region (POISSON).

These are covered in STAT 225. Here are some results you will be able to perform by the end of *that* course.

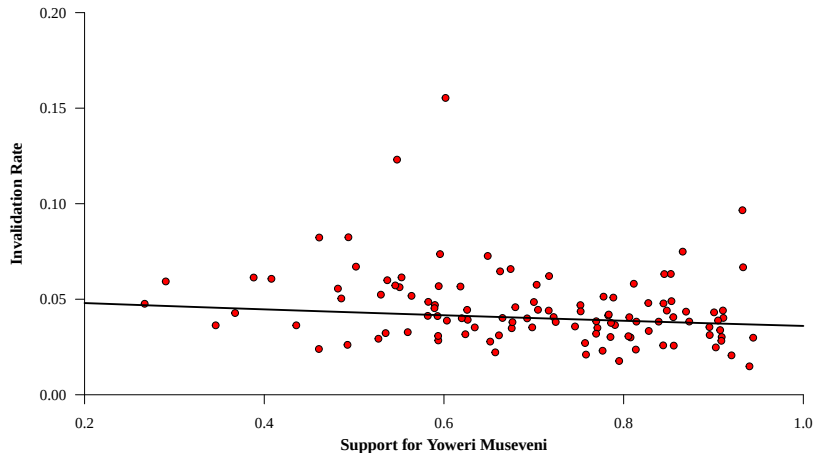
Dichotomous (BERNOULLI) Response

The following is a graphic of the relationship between a homeowner's age and the likelihood that the homeowner has purchased flood insurance.



BINOMIAL Response

The following is a graphic of the relationship between a vote and the likelihood it was invalidated in the 2011 presidential election in Uganda.



Count (Rate) Response

The following is a graphic estimating the number of citizen's initiatives in states based on their population and geographic region (for states that allow the initiative).

